

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-02

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TOPICS TO BE COVERED

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T- Functions as Circular Functions

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Related Problems

EXERCISE – 3.1

QUESTION

Find the **radian measures** corresponding to the following **degree measures**:

- (i) 25°
- (ii) $-47^\circ 30'$
- (iii) 240°
- (iv) 520°
- (i) $25^\circ = \frac{5}{36} \times \frac{\pi}{180}$
- $\Rightarrow 25^\circ = \frac{5\pi}{36} \text{ radians.}$
- (ii) $-47^\circ 30' = 47^\circ + 30'$
- $= 47^\circ + 30' \times \frac{1}{60}$
- $= 47^\circ + \frac{1}{2}^\circ$

$= \left(\frac{95}{2}\right)^\circ$

$= \frac{19}{2} \times \frac{\pi}{180}$

$= \frac{19\pi}{72} \text{ radians}$

$60' = 1^\circ$

$\Rightarrow 1' = \left(\frac{1}{60}\right)^\circ$

$1^\circ = \frac{\pi}{180} \text{ radians}$

$1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$

QUESTION

$$1^\circ = 60'$$

Find the degree measures corresponding to the following radian measures

(Use $\pi = \frac{22}{7}$)

(i) $\frac{11}{16}$

(ii) -4

(iii) $\frac{5\pi}{3}$

(iv) $\frac{7\pi}{6}$

$$\begin{aligned} \text{(i) } \frac{11}{16} \text{ radians} &= \frac{11}{16} \times \frac{180}{\pi} \\ &= \frac{11}{16} \times \frac{180 \times 7}{22} \\ &= \frac{45 \times 7}{8} = \frac{315}{8} \end{aligned}$$

$$\begin{aligned} &= 39^\circ + \left(\frac{3}{8}\right)^\circ \\ &= 39^\circ + \left(\frac{3}{8} \times 60\right)' \\ &= 39^\circ + \left(\frac{45}{2}\right)' \\ &= 39^\circ + \left(22 + \frac{1}{2}\right)' \\ &= 39^\circ + 22' + \left(\frac{1}{2} \times 60\right)'' \\ &= 39^\circ 22' 30'' \end{aligned}$$

$$\begin{array}{r} 39 \\ 8 \overline{) 315} \\ \underline{24} \\ 75 \\ \underline{72} \\ 3 \end{array}$$
$$1' = 60''$$

QUESTION

A wheel makes 360 revolutions in one minute. Through how many radians does it turn in one second?

$$12\pi.$$

$$360^\circ = 2\pi$$

$$360 \rightarrow 60 \text{ seconds}$$

$$2\pi \times 6 = 12\pi.$$

$$\Rightarrow 1 \text{ Sec} \rightarrow \frac{360}{60} = 6$$

QUESTION

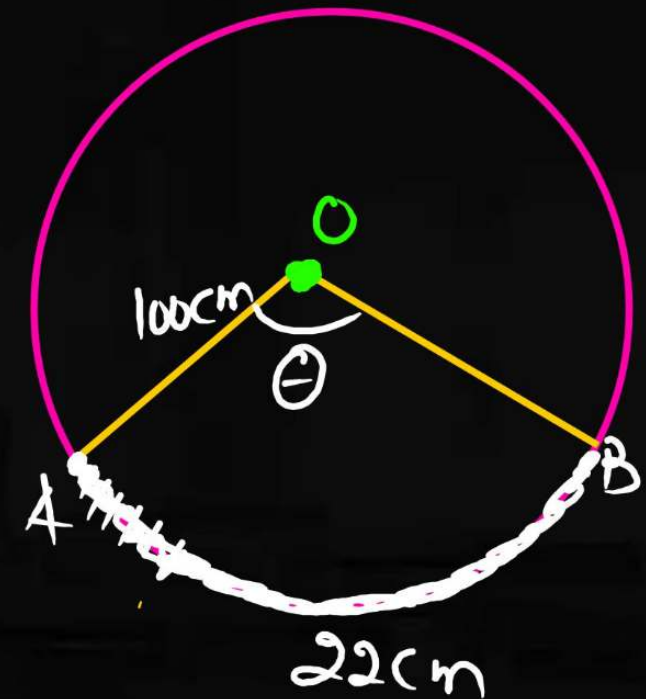
Find the degree measure of the angle subtended at the centre of a circle of radius 100 cm by an arc of length 22 cm

$$\text{(Use } \pi = \frac{22}{7} \text{)}$$

$$\begin{aligned} \text{(Sol)} \text{ We know that } \theta &= \frac{l}{r} \\ \Rightarrow \theta &= \frac{22}{100} = \frac{11}{50} \text{ radians} \end{aligned}$$

$$\Rightarrow \theta = \frac{11}{50} \times \frac{180}{\pi} = \frac{11}{5} \times \frac{18}{7} \times 7$$

$$\begin{aligned} \Rightarrow \theta &= \left(\frac{63}{5} \right)^{\circ} \\ \Rightarrow \theta &= 12^{\circ} + \left(\frac{3}{5} \right)^{\circ} \\ \Rightarrow \theta &= 12^{\circ} + \frac{3}{5} \times 60' \\ \Rightarrow \theta &= 12^{\circ} 36' \end{aligned}$$



QUESTION

In a circle of diameter 40 cm, the length of a chord is 20 cm. Find the length of minor arc of the chord.

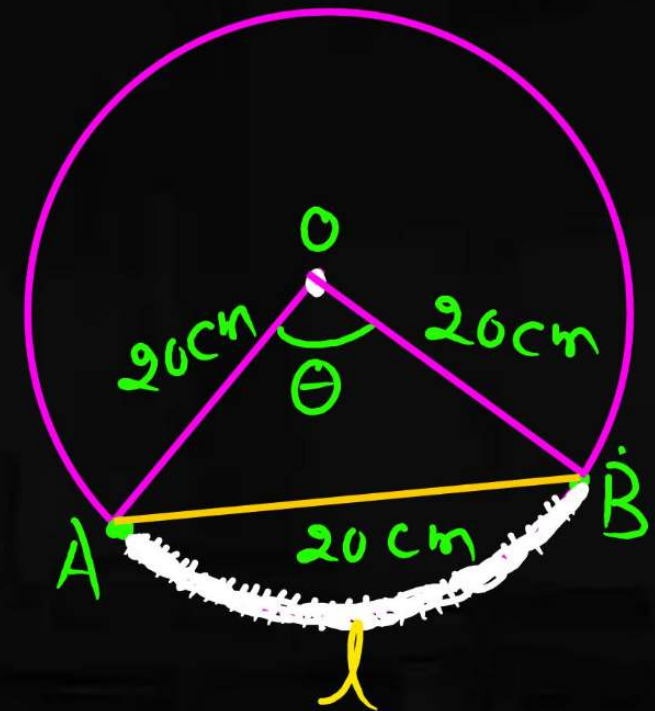
(Sol) Here $\theta = 60^\circ = 60 \times \frac{\pi}{180}$

$$\Rightarrow \theta = \frac{\pi}{3}$$

$$\Rightarrow \frac{l}{r} = \frac{\pi}{3}$$

$$\Rightarrow l = \frac{\pi}{3} \times 20$$

$$\Rightarrow l = \frac{20\pi}{3}$$



QUESTION

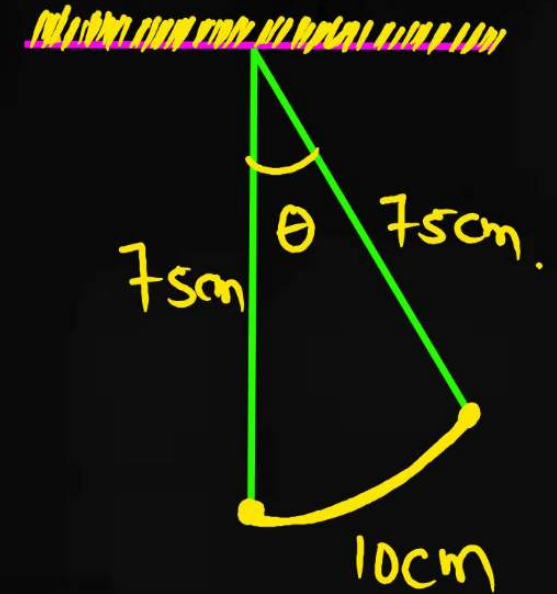
If in two circles, arcs of the same length subtend angles 60° and 75° at the centre, find the ratio of their radii.

(Homework)

QUESTION

Find the angle in radian through which a pendulum swings if its length is 75 cm and the tip describes an arc of length.

- (i) 10 cm
- (ii) 15 cm
- (iii) 21 cm



(i)
(Sol) Here $\theta = \frac{10}{75} = \frac{2}{15}$ radians

T-Functions Defined as Circular Functions

- Trigonometric functions are known as circular functions because they can be defined with the help of a unit circle.

In rt. $\triangle OPM$,

$$\cos x = \frac{OM}{OP}$$

$$\Rightarrow OM = OP \cos x$$

$$\Rightarrow OM = \cos x$$

$$\Rightarrow a = \cos x$$

$$\text{Also, } \sin x = \frac{PM}{OP}$$

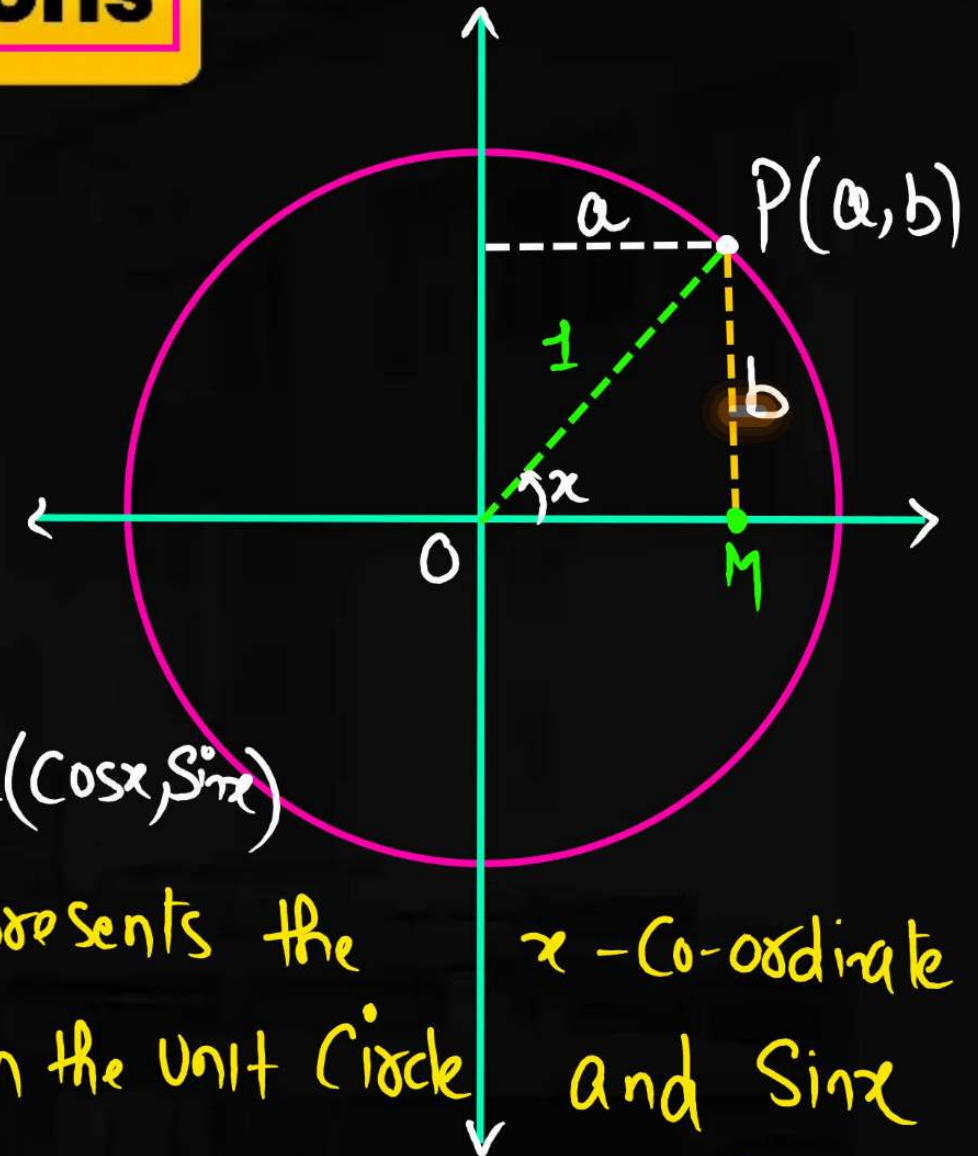
$$\Rightarrow PM = OP \sin x$$

$$\Rightarrow PM = \sin x$$

$$\Rightarrow b = \sin x$$

$$\text{Therefore } (a, b) = (\cos x, \sin x)$$

Thus $\cos x$ represents the x -Co-ordinate of any point on the unit circle and $\sin x$ represents the y -Co-ordinate of any point on the unit circle.

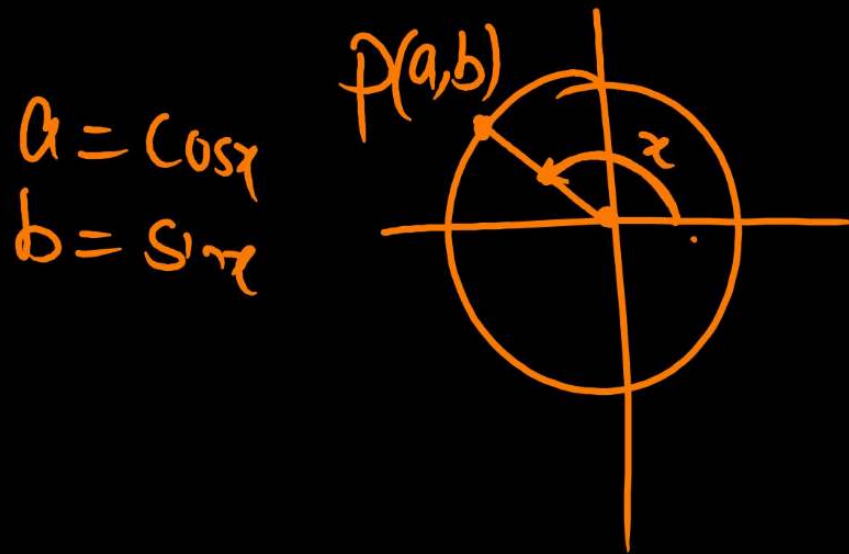
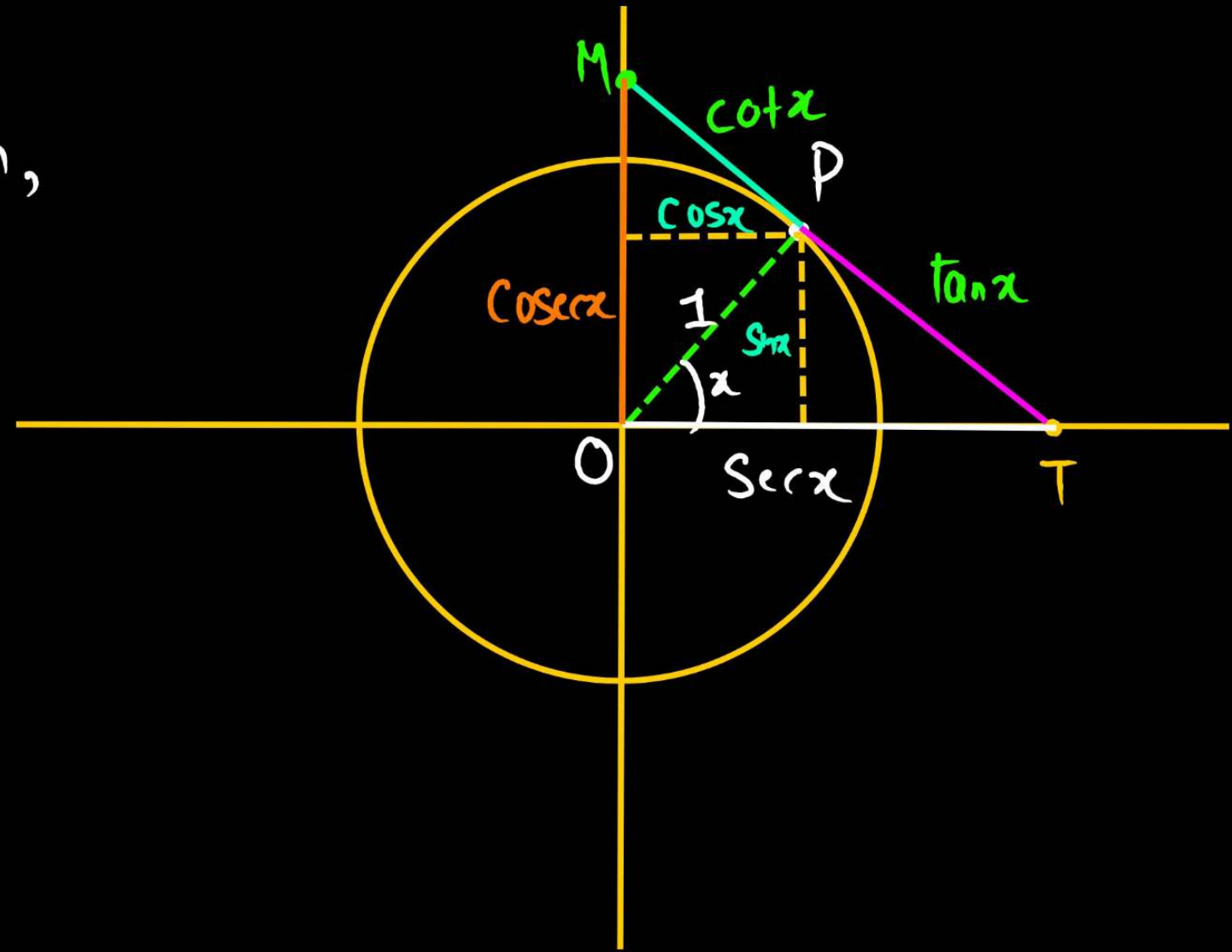


Also, By using Pythagoras theorem,

$$OP^2 = OM^2 + PM^2.$$

$$\Rightarrow a^2 + b^2 = 1^2$$

$$\Rightarrow \boxed{\cos^2 x + \sin^2 x = 1}$$



$$a = \cos x$$

$$b = \sin x$$

Sign of Trigonometric Functions

In 1st Quadrant:

All T-functions are positive.

In 2nd Quadrant:

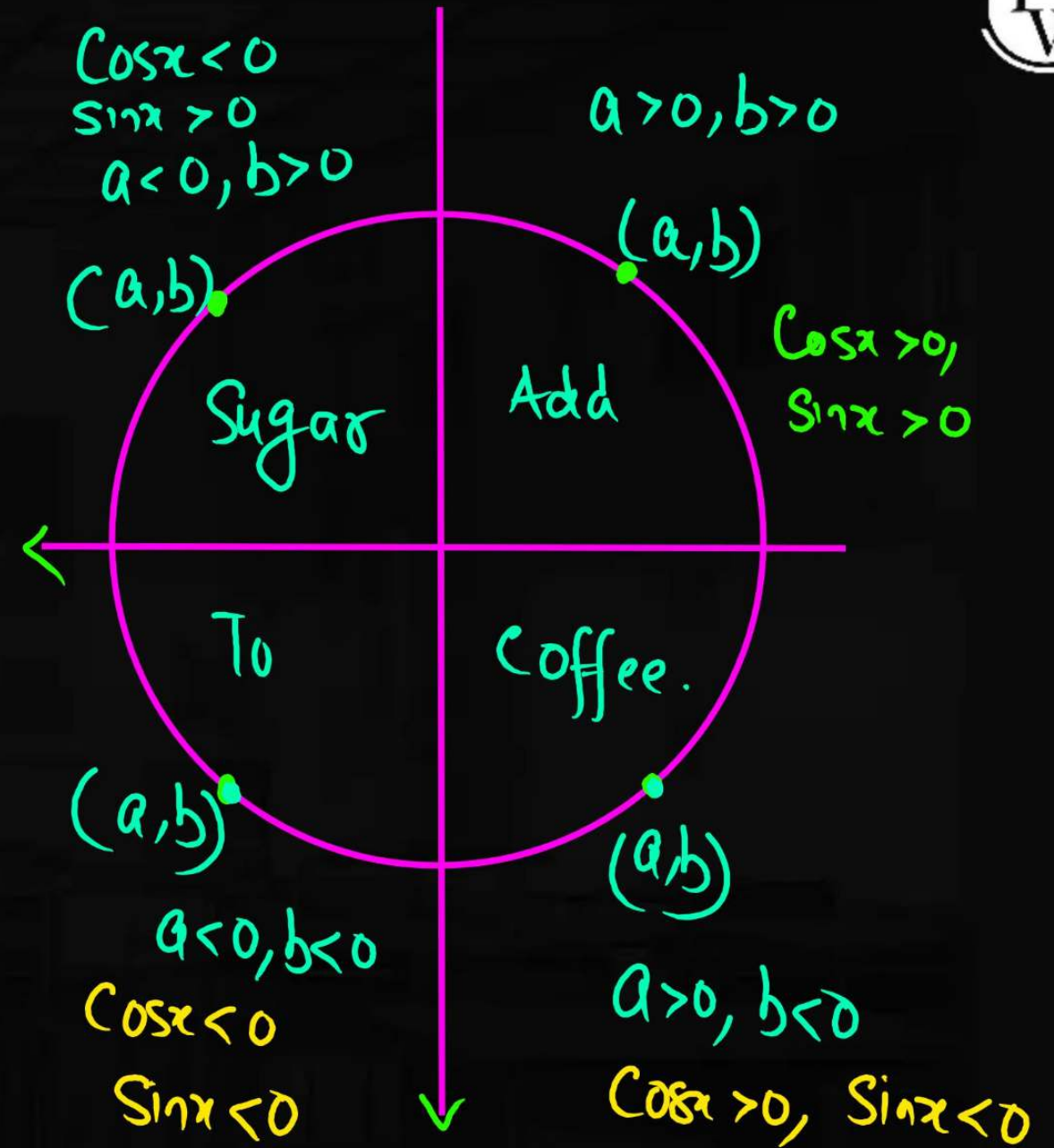
only $\sin x$ and $\operatorname{cosec} x$ are positive

In 3rd Quadrant:

only $\tan x$ and $\cot x$ are positive.

In 4th Quadrant:

only $\cos x$ and $\sec x$ are positive.



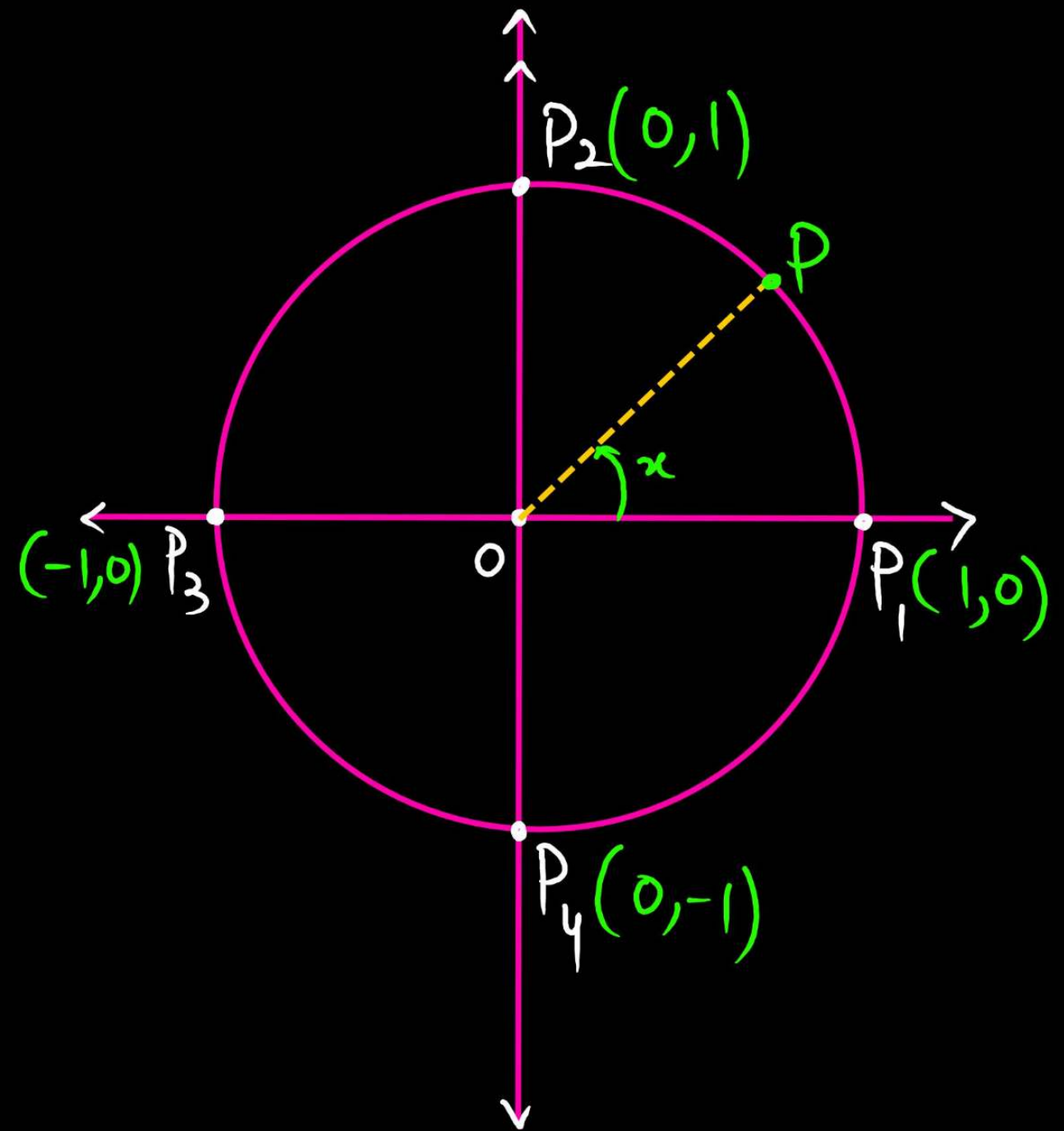
$$* \cos 0 = 1, \sin 0 = 0$$

$$* \cos \frac{\pi}{2} = 0, \sin \frac{\pi}{2} = 1$$

$$* \cos(\pi) = -1, \sin \pi = 0$$

$$* \cos\left(\frac{3\pi}{2}\right) = 0, \sin\left(\frac{3\pi}{2}\right) = -1$$

$$* \cos\left(\frac{(2n+1)\pi}{2}\right) = 0, \sin n\pi = 0, n \in \mathbb{Z}$$



Quadrant	I	II	III	IV
$\sin x$	+	+	-	-
$\cos x$	+	-	-	+
$\tan x$	+	-	+	-
$\operatorname{cosec} x$	+	+	-	-
$\sec x$	+	-	-	+
$\cot x$	+	-	+	-

QUESTION

The Domain and range of $\sin x$ are respectively equal to

- ☒ 1 $[-1,1]$ and $\mathbb{R}$
- ☒ 2 $\mathbb{R}$ and $[0,1]$
- ☒ 3 $\mathbb{R}$ and $[-1,1]$
- ☒ 4 $[-1,0]$ and $\mathbb{R}$

QUESTION

The Domain and range of $\cos x$ are respectively equal to

1 $[-1,1]$ and $\mathbb{R}$ ✗

2 $\mathbb{R}$ and $[0,1]$ ✗

3 $\mathbb{R}$ and $[-1,1]$ ✓

4 $[-1,0]$ and $\mathbb{R}$ ✗

QUESTION

$$\cos 90^\circ = 0$$

The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 179^\circ$ is

☒ 1 $\frac{1}{\sqrt{2}}$

☒ 2 0

☒ 3 1

☒ 4 -1

$$\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ \cos 91^\circ \dots \cos 179^\circ = 0$$

QUESTION

If $5\sin\theta = 3$, then $\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta}$ is equal to :

☒ 1 $\frac{1}{4}$

☒ 2 4

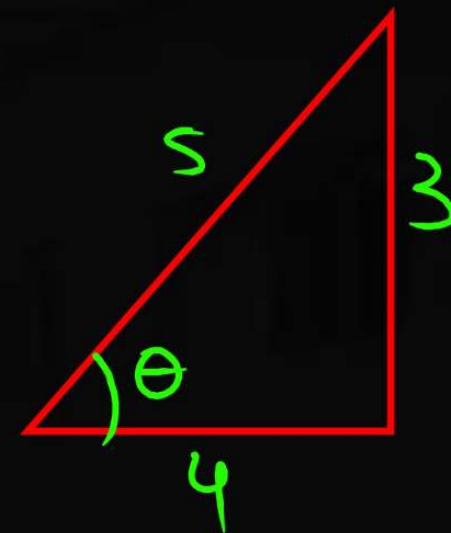
☐ 3 2

☐ 4 None of these

(Sol) Given that $5\sin\theta = 3$

$\Rightarrow \sin\theta = \frac{3}{5}$

Therefore
$$\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta} = \frac{\left(\frac{5}{4} + \frac{3}{4}\right)}{\left(\frac{5}{4} - \frac{3}{4}\right)}$$
$$= \frac{\left(\frac{8}{4}\right)}{\left(\frac{2}{4}\right)} = \frac{2}{2} = 4$$



QUESTION

If $\tan \theta = 3$ and θ lies in third quadrant, then the value of $\sin \theta$ is

~~1~~ $\frac{1}{\sqrt{10}}$

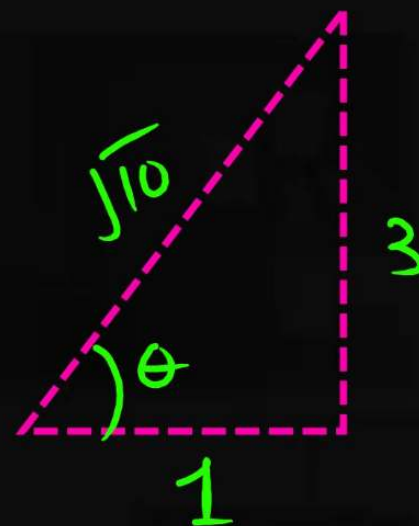
~~2~~ $-\frac{1}{\sqrt{10}}$

☒ 3 $-\frac{3}{\sqrt{10}}$

~~4~~ $\frac{3}{\sqrt{10}}$

$$\tan \theta = 3$$

$$\text{So, } \sin \theta = -\frac{3}{\sqrt{10}}$$



QUESTION

If $2 \cos^2 \theta + 5 \sin \theta - 4 = 0$, then the value of $16 \sin^4 \theta + 4 \cos^2 \theta$ is equal to :

~~1~~

8

~~2~~

7

~~3~~

4

~~4~~

3

(Sol) we have

$$2 \cos^2 \theta + 5 \sin \theta - 4 = 0$$

$$\Rightarrow 2(1 - \sin^2 \theta) + 5 \sin \theta - 4 = 0$$

$$\Rightarrow 2 - 2 \sin^2 \theta + 5 \sin \theta - 4 = 0$$

$$\Rightarrow -2 \sin^2 \theta + 5 \sin \theta - 2 = 0$$

$$\Rightarrow 2 \sin^2 \theta - 5 \sin \theta + 2 = 0$$

Put $y = \sin \theta$, we get;

$$2y^2 - 5y + 2 = 0$$

$$\Rightarrow 2y^2 - 4y - y + 2 = 0$$

$$\Rightarrow 2y(y - 2) - 1(y - 2) = 0$$

$$\Rightarrow (y - 2)(2y - 1) = 0$$

$$\Rightarrow (\sin \theta - 2)(2 \sin \theta - 1) = 0$$

$$\Rightarrow 2 \sin \theta - 1 = 0, (\sin \theta - 2 \neq 0)$$

$$\Rightarrow \boxed{\sin \theta = \frac{1}{2}}$$

$$\text{So, } \cos^2 \theta = 1 - \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta = 1 - \frac{1}{4}$$

$$\Rightarrow \boxed{\cos^2 \theta = \frac{3}{4}}$$

$$\begin{aligned} \text{Therefore } 16 \sin^4 \theta + 4 \cos^2 \theta \\ = 16 \times \left(\frac{1}{2}\right)^4 + 4 \left(\frac{3}{4}\right) = \cancel{16} \times \frac{1}{16} + 3 \\ = 4. \end{aligned}$$

QUESTION (MA Q)

If $\sin^2 \theta + 3 \cos \theta - 2 = 0$, then $\cos^3 \theta + \sec^3 \theta$ is equal to

☒ 18

☐ 9

☐ 4

☐ $\frac{1}{4}$

$$(Sol) \sin^2 \theta + 3 \cos \theta - 2 = 0$$

$$\Rightarrow 2 - \sin^2 \theta = 3 \cos \theta$$

$$\Rightarrow 2 - (1 - \cos^2 \theta) = 3 \cos \theta$$

$$\Rightarrow 2 - 1 + \cos^2 \theta = 3 \cos \theta$$

$$\Rightarrow 1 + \cos^2 \theta = 3 \cos \theta$$

$$\Rightarrow \frac{1}{\cos \theta} + \frac{\cos^2 \theta}{\cos \theta} = 3$$

$$\Rightarrow \sec \theta + \cos \theta = 3 \quad \text{--- (1)}$$

Cubing on b/s, we get,

$$(\sec \theta + \cos \theta)^3 = 27$$

$$\Rightarrow \sec^3 \theta + \cos^3 \theta + 3 \sec \theta \cos \theta (\sec \theta + \cos \theta) = 27$$

$$\Rightarrow \cos^3 \theta + \sec^3 \theta + \frac{3}{\cos \theta} \times \cos \theta (3) = 27$$

$$\Rightarrow \cos^3 \theta + \sec^3 \theta + 9 = 27$$

$$\Rightarrow \cos^3 \theta + \sec^3 \theta = 18$$

Thank
You